FROM DATA TO DECISIONS

An end-to-end Technical Business Analysis and Operations Intelligence demonstration: data validation, SQL investigation, interactive analytics, requirements and a controlled pilot decision.

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SIMULATED REPORTING: JUN 2025 - AUG 2026

Investigate a sustained SLA decline

A fictional customer-operations team sees weaker SLA performance even though contact volume remains stable. The stakeholder asks: what changed, where should we investigate, and what would a controlled response look like?

VALID CONTACTS

78,000

15 simulated months

BASELINE SLA

88.3%

Jun-Aug 2025

DIAGNOSTIC SLA

82.9%

Jan-Mar 2026

FOLLOW-UP SLA

82.8%

Jun-Aug 2026

WHAT THE DATA DEMONSTRATES

The simulated Jan-Mar 2026 SLA declined by 5.4 percentage points versus Jun-Aug 2025. Through Aug 2026, service levels remain weak: the latest quarter is 82.8%.

DELIVERABLES

- Governed contact-level dataset and quality checks
- Cohort-level SQL investigation and 15-month trend
- Interactive dashboard, what-if scenario and UAT matrix
- Prioritised improvement pilot with explicit stop rules

Trust the dataset before trusting the dashboard

Generated operational data are subjected to explicit checks. Invalid fictional records are quarantined before any published KPI is calculated.

LAND
78,072
source records

→

CHECK
72
invalid rows

→

CURATE
78,000
accepted contacts

→

MART
360
aggregated groups

QUARANTINE REASON



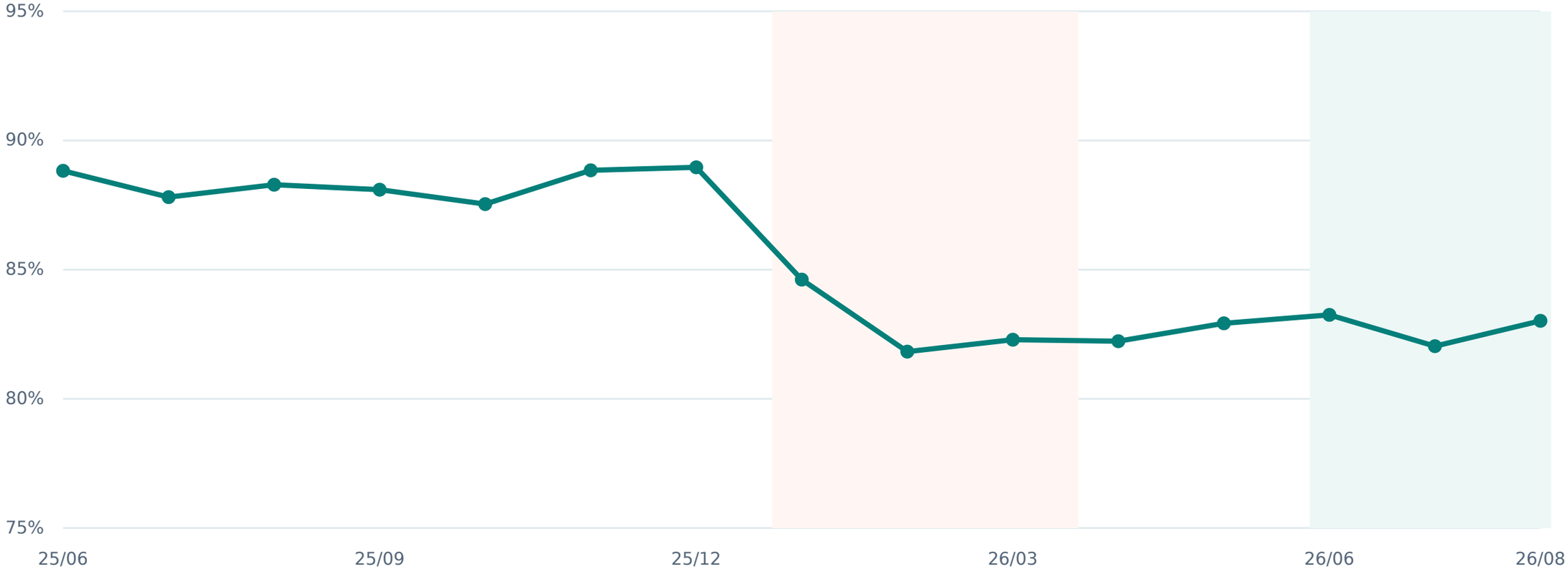
REPORTING CONTRACT

Grain: one fictional contact. SLA/FCR: all valid contacts. CSAT: only surveyed respondents. Exception handling: raw-to-curated reconciliation, ID uniqueness and field validity.

9 of 9 updated automated checks passed

15 months of SLA performance

The original Jan-Mar 2026 diagnostic remains visible, and five additional synthetic months extend the reporting window to August 2026. No improvement is attributed to an actual intervention.



INTERPRETATION

The prolonged weak period is a simulated diagnostic finding. It does not demonstrate a deployed solution. The follow-up quarter (Jun-Aug 2026) averages 82.8% SLA, below the 88.3% baseline.

Separate service deterioration from contact mix

First compare the three-month baseline against the Jan-Mar 2026 deterioration. Then check the later synthetic follow-up without implying causality.

WITHIN-QUEUE EFFECT

-5.38 pp

Mix-adjusted, baseline to diagnostic

CONTACT MIX EFFECT

-0.02 pp

Team/channel distribution change

TOTAL CHANGE

-5.40 pp

Components sum to the net decline

BILLING / EMAIL SLA COMPARISON

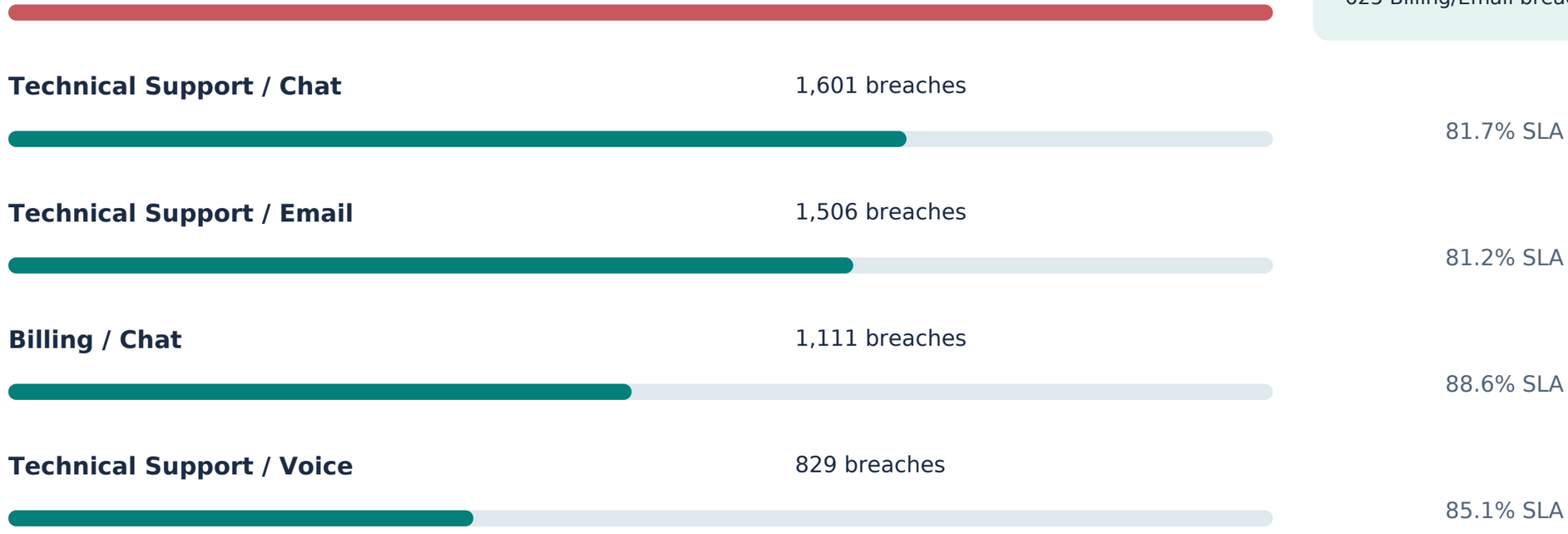


Interpretive caution: this pattern was deliberately generated and does not prove staffing, routing, case complexity or any specific operational cause. Validate hypotheses with real process owners and appropriate data before taking action.

Prioritise by breaches, not only percentages

A low SLA rate in a tiny queue may matter less than a moderately weak high-volume segment. This ranking uses all 15 months of fictional contact data.

HIGHEST SYNTHETIC CONTRIBUTIONS TO SLA BREACHES



FOLLOW-UP COHORT

1,866 contacts

625 Billing/Email breaches

Focus next on queue routing, rework, staffing, backlog and case mix - possible explanations, not observed root causes. The cohort and breach counts reconcile to the synthetic dataset.

Turn an insight into a testable scenario

The what-if model illustrates a possible SLA arithmetic effect. It does not estimate a causal benefit or represent a real intervention.

DIAGNOSTIC COHORT

1,827

Billing / Email, Jan-Mar 2026

SLA BREACHES

662

Same diagnostic cohort

25% ASSUMPTION

166

Hypothetical avoided breaches

PERIOD-WIDE LIFT

+0.21 pp

If those cases met SLA over 78k

MODEL GOVERNANCE

A logistic intake-risk baseline using the original Jun 2025-Mar 2026 synthetic subset achieved a chronological holdout ROC AUC of 0.66 (36,400 train / 15,600 test). This indicates modest discrimination; the model was not retrained or evaluated on April-August 2026.

BUSINESS DECISION

- Use the scenario to estimate scale, not expected uplift.
- Agree cohort, controls and operational limits before a pilot.
- Track SLA, FCR and CSAT to avoid metric gaming.
- Re-evaluate and document the result before scaling.

One traceable path from requirement to validated output

SQL and Python evidence is provided in the accompanying website ZIP. These are synthetic automated tests, not stakeholder acceptance or employer UAT sign-off.

REQ-04 / QUERY REPRODUCIBLE IN SQLITE

```
SELECT COUNT(*) AS contacts,
       SUM(CASE WHEN sla_met=0
           THEN 1 ELSE 0 END) AS breaches,
       ROUND(100.0*AVG(sla_met),1) AS sla_pct
FROM contacts
WHERE team='Billing' AND channel='Email'
  AND month BETWEEN '2026-01' AND '2026-03';
```

SQL OUTPUT

1,827 contacts

662 breaches

63.8% SLA

9/9 checks passed: counts, quarantine, uniqueness, SLA denominator, cohort result, 25% model, zero-recovery, end-month and decomposition.

BUSINESS REQUIREMENT TO ACCEPTANCE TEST

REQ-01	**Reject invalid records**	72 deliberately planted exceptions excluded
REQ-02	**Reconcile KPI definition**	Filtered totals and SLA denominator agree
REQ-03	**Explain variation**	Mix + within effects equal net difference
REQ-04	**Quantify priority queue**	SQL cohort matches dashboard

A pilot with an explicit decision rule

No intervention actually took place. This section proposes a six-week evaluation to test whether routing and ageing-case triage could improve the high-breach queue.

PROPOSED SIX-WEEK SEQUENCE

WEEKS 1-2	Agree KPI definitions, cohort and matched comparison
WEEKS 3-6	Trial routing/triage; record exceptions each day
WEEKLY	Review SLA, backlog, FCR, surveyed CSAT and mix
FINAL	Review adjusted outcomes and sample adequacy

ILLUSTRATIVE GATES

GO / EXPAND

Consider expanding only if adjusted SLA improves by at least 5 percentage points versus a valid comparator, without meaningful FCR or CSAT decline.

STOP / REDESIGN

Stop if customer outcomes deteriorate, the comparison is unreliable or the necessary data controls are missing. Agree gates and sample size with the sponsor first.

DISCLOSURE: SELF-DIRECTED SYNTHETIC CASE STUDY. REPORTING DATES ARE NOT PROJECT-COMPLETION DATES.